

TELUGU MOVIE DATA ANALYSIS USING HIVE AND R

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Certificate

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This is to certify that the work present in this Project entitled “**TELUGU MOVIE DATA ANALYSIS USING HIVE AND R**” has been carried out by **M. Sree Annapurna, B. Jyothirmai, G. Madhurya, M. Nandita, M. Veni, and N. Vamsi** to **Dr. Saleti Sumalatha** under my supervision. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of Bachelor of Science in the **School of Engineering and Sciences**.

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Abstract

This project provides a detailed analysis of Telugu movie data, utilizing big data analytics to gain insights into the film industry. It covers data collection, preprocessing, and exploratory analysis of a dataset containing features such as movie titles, release years, genres, certifications, runtimes, and ratings. Through advanced data processing and statistical techniques, the study identifies key patterns in production trends, audience preferences, and content characteristics.

Using R programming for statistical analysis and data visualization, the project reveals insights into metrics like movie ratings, genre distribution, runtime variation, and release patterns. Significant trends, such as changes in genre popularity and factors influencing movie success, are highlighted, offering a deeper understanding of the Telugu film industry.

This analysis underscores the potential of big data in exploring entertainment industry dynamics, providing valuable implications for filmmakers, producers, and researchers. The study's findings serve as a foundation for further research and encourage data-driven decision-making within the Telugu cinema landscape.

Introduction

Movies are a significant part of entertainment, and understanding their success metrics and characteristics is crucial for producers, directors, and audiences alike. The "Preprocessed_TeluguMovies_Dataset.csv" dataset offers a rich source of information about Telugu movies. The dataset includes various attributes, such as runtime, genre, and ratings, which provide an opportunity to explore trends in the industry.

This project analyses Telugu movie data to gain insights into trends, audience preferences, and factors influencing success in the Tollywood film industry. Leveraging a comprehensive dataset with features like movie titles, genres, certifications, runtimes, and ratings, the study aims to uncover key patterns within the industry.

Advanced data processing tools like Hive and R programming are utilized for statistical analysis, while Python-based visualization in Google Collab effectively presents findings. This analysis highlights genre popularity, runtime trends, and rating distributions, providing valuable insights for filmmakers, producers, and researchers. By integrating big data analytics and visualization, the project underscores the importance of data-driven decision-making within the Telugu movie landscape.

This study employs Hive for querying the dataset efficiently, enabling scalable analysis of large movie datasets. Preprocessing ensures that the dataset is clean and consistent for analysis, which is followed by executing 15 Hive queries. These queries aim to answer questions such as the most common genres, top-rated movies, certification trends, and runtime distributions. Furthermore, 10 visualizations are created to provide intuitive insights into the data, making it easier to interpret complex relationships and trends.

By leveraging data analytics and visualization techniques, this project aims to understand the Telugu movie industry's dynamics, providing insights into audience preferences and the characteristics of successful films.

1.1 Motivation

The Telugu film industry, a cornerstone of Indian cinema, has seen tremendous growth and evolution over the decades, captivating audiences not only in India but also among the global diaspora. With its rich storytelling, vibrant performances, and innovative filmmaking techniques, Tollywood has become a significant player in the entertainment industry. This growth has generated a wealth of data that reflects the changing trends in audience preferences, genre popularity, and the factors influencing movie success.

Understanding the dynamics of the Telugu film industry through data analysis offers immense value to filmmakers, producers, and researchers. The availability of large datasets capturing various aspects of movies, including genres, certifications, runtimes, ratings, and audience responses, creates an opportunity to uncover hidden patterns and trends. Such insights can inform strategic decisions in content creation, marketing, and distribution, ultimately enhancing audience engagement and satisfaction.

The motivation behind this project stems from the increasing importance of data driven decision-making in creative industries. By leveraging advanced data analytics techniques, we aim to bridge the gap between audience expectations and cinematic offerings, enabling a more profound understanding of what resonates with viewers. This analysis not only helps filmmakers optimize their content but also contributes to academic research in film studies and the broader domain of entertainment analytics.

In a rapidly evolving digital landscape, where audience preferences are constantly shifting, this project aspires to provide meaningful insights into the Telugu film industry's trajectory, helping stakeholders stay ahead in an increasingly competitive market.

1.2 Required Environments

1. Collab Environment: Python development and Preprocessing
2. Oracle DM Virtual Box: To run Horton works Dockers
Sandbox: Runs Ambari Admin, SSH Client.

Problem Survey

1. The goal of this research is to utilize big data techniques to conduct a comprehensive problem survey of Telugu movie data analysis. The objective is to analyze a substantial dataset from the Telugu film industry to uncover trends, identify challenges, and gain meaningful insights.
2. The Telugu film industry, one of the largest regional film industries in India, generates a wealth of data annually. This data includes movie titles, release years, genres, certifications, runtimes, ratings, and audience reviews. By leveraging big data techniques, we aim to extract valuable insights from this dataset, such as trends in movie production, genre popularity, audience preferences, and factors influencing movie success.
3. The primary goal of this project is to explore key challenges and opportunities within the Telugu film industry. Specific areas of investigation include identifying the most popular genres, examining runtime trends over the years, analyzing rating distributions to evaluate audience feedback, and understanding the impact of certifications on movie performance.
4. This project's main aim is to perform a problem survey to identify key questions or challenges in Telugu movie analysis. For instance, what makes certain genres more successful than others? How have audience preferences evolved over time? What is the relationship between runtime and movie ratings? By addressing these and similar questions, we hope to gain deeper insights into the dynamics of the Telugu film industry.
5. Ultimately, this research aspires to enhance our understanding of the Telugu film landscape by uncovering significant patterns and trends in the data. These findings will not only support data-driven decision-making for filmmakers and producers but also serve as a foundation for further research in the field of movie analytics.

Dataset Description

2.1 About Dataset

The dataset contains 1400 entries and 9 attributes, providing comprehensive information about Telugu movies

Columns:

- S.no
- Movie
- Year
- Certificate
- Genre
- Overview
- Runtime
- Rating
- Movie Ratings

2.2 Description of Dataset Attributes:

Here is a detailed description:

Attributes:

1. **S.no:** A unique identifier for each movie (integer).
2. **Movie:** The name of the movie (string).
3. **Year:** The year of release (integer, ranging from 1932 to 2020).
4. **Certificate:** The certification of the movie (e.g., U, UA, A, etc., with 8 unique categories).
5. **Genre:** The genre(s) of the movie, which may include combinations such as Action, Drama, Thriller, etc. (171 unique genres).
6. **Overview:** A short description or summary of the movie. Some entries have "No overview available" as a placeholder.
7. **Runtime:** The duration of the movie in minutes (integer, ranging from 0 to 300).
8. **Rating:** IMDb-style user rating (float, ranging from 1.0 to 9.5).
9. **Movie Ratings:** The number of ratings received (integer, ranging from 49 to 99,114).
 - The dataset includes diverse movie genres, certificates, and runtimes, capturing a wide span of Telugu cinema from 1932 to 2020.
 - The dataset is rich in variability, with genres, ratings, and runtime providing opportunities for exploratory analysis, trend observation, and predictive modelling.

➤ Pre-processing:

- This Python code snippet is used in Google Colab to work with external files:

1. `Drive.mount ()` mounts the Google Drive to access files stored in it.
2. `Files.upload ()` allows uploading files (e.g., `TeluguMovies_dataset.csv`) from your local system to Colab for further processing.

```
[ ] from google.colab import drive
    drive.mount('/content/drive', force_remount=True)
```

Mounted at /content/drive

```
[ ] from google.colab import files
    uploaded = files.upload()
```

Choose Files TeluguMovi..._dataset.csv

- **TeluguMovies_dataset.csv**(text/csv) - 292441 bytes, last modified: 11/14/2024 - 100% done

Saving TeluguMovies_dataset.csv to TeluguMovies_dataset.csv

- The dataset is read into a pandas Data Frame using the ``read_csv ()`` function, allowing for data manipulation and analysis in Python

```
[ ] import pandas as pd
    df = pd.read_csv("TeluguMovies_dataset.csv")
```

df.head()

	S.no	Movie	Year	Certificate	Genre	Overview	Runtime	Rating	Movie_Ratings
0	0	Bahubali: The Beginning	2015.0	UA	Action, Drama	In ancient India, an adventurous and darin...	159	8.1	99114
1	1	Baahubali 2: The Conclusion	2017.0	UA	Action, Drama	When Shiva, the son of Bahubali, learns ab...	167	8.2	71458
2	2	1 - Nenokkadine	2014.0	UA	Action, Thriller	A rock star must overcome his psychologica...	170	8.1	42372
3	3	Dhoom:3	2013.0	UA	Action, Thriller	When Sahir, a circus entertainer trained i...	172	5.4	42112
4	4	Ra.One	2011.0	U	Action, Adventure, Sci-Fi	When the titular antagonist of an action g...	156	4.6	37211

- The code snippet reads a dataset of 1,400 Telugu movies into a pandas DataFrame df, displaying attributes like title, year, genre, runtime, and rating

✓ [14] df

	S.no	Movie	Year	Certificate	Genre	Overview	Runtime	Rating	Movie_Ratings
0	0	Bahubali: The Beginning	2015	UA	Action, Drama	In ancient India, an adventurous and daring ma...	159	8.1	99114
1	1	Baahubali 2: The Conclusion	2017	UA	Action, Drama	When Shiva, the son of Bahubali, learns about ...	167	8.2	71458
2	2	1 - Nenokkadine	2014	UA	Action, Thriller	A rock star must overcome his psychological in...	170	8.1	42372
3	3	Dhoom:3	2013	UA	Action, Thriller	When Sahir, a circus entertainer trained in ma...	172	5.4	42112
4	4	Ra.One	2011	U	Action, Adventure, Sci-Fi	When the titular antagonist of an action game ...	156	4.6	37211
...	...	...	...	...	...	...	...	...	...
1395	1395	Maro Monagadu	1985	UNKNOWN	Unknown	No overview available	0	8.6	49
1396	1396	Jakkanna	2016	UNKNOWN	Comedy, Drama	The movie is about an attempt by Sunil the pro...	0	6.3	49
1397	1397	Muvva Gopaludu	1987	UNKNOWN	Drama, Romance	Muvva Gopaludu is a 1987 Indian Telugu film, D...	137	7.8	49
1398	1398	Ninney Ishta Paddaanu	2003	U	Unknown	Hero Charan (Tarun) a middle class family guy,...	0	5.9	49
1399	1399	9 Nelalu	2001	U	Drama	Surendra marries Savitri, an orphan. In a turn...	0	7.3	49

1400 rows x 9 columns

- The code calculates and prints the count of missing values for each column in the DataFrame df. The output shows the columns with missing values (e.g., Certificate has 449 missing values, Overview has 179), while others like S.no and Movie have none.

```
[ ] # Check for missing values
missing_values = df.isnull().sum()
print("Missing values per column:\n", missing_values)
```

```
Missing values per column:
S.no          0
Movie         0
Year         48
Certificate   449
Genre        11
Overview     179
Runtime       0
Rating        0
Movie_Ratings 0
dtype: int64
```

- The code snippet ensures data consistency by converting columns to specific data types (`int`, `float`, or `str`). It also cleans string columns by stripping whitespace and standardizing case (e.g., title case for `Movie` and `Genre`, uppercase for `Certificate`).

```

▶ # Convert data types for consistency
df['Year'] = df['Year'].astype(int)
df['Runtime'] = df['Runtime'].astype(int)
df['Rating'] = df['Rating'].astype(float)
df['Movie'] = df['Movie'].astype(str)
df['Certificate'] = df['Certificate'].astype(str)
df['Genre'] = df['Genre'].astype(str)
df['Overview'] = df['Overview'].astype(str)

```

```

[ ] # Strip whitespace and standardize the case for string columns
df['Movie'] = df['Movie'].str.strip().str.title()
df['Genre'] = df['Genre'].str.strip().str.title()
df['Certificate'] = df['Certificate'].str.strip().str.upper()
df['Overview'] = df['Overview'].str.strip()

```

- The code snippet fills missing values in specific columns of a DataFrame (`df`) using different strategies:
 1. The `Year` column's missing values are replaced with the column's median value.
 2. The `Certificate` and `Genre` columns' missing values are replaced with the string `"Unknown"`.
 3. The `Overview` column's missing values are replaced with the string `"No overview available"`.

```

▶ # Fill missing values in 'Year' with the median year
df.loc[:, 'Year'] = df['Year'].fillna(df['Year'].median())

# Fill missing values in 'Certificate' and 'Genre' with 'Unknown'
df.loc[:, 'Certificate'] = df['Certificate'].fillna('Unknown')
df.loc[:, 'Genre'] = df['Genre'].fillna('Unknown')

# Fill missing values in 'Overview' with 'No overview available'
df.loc[:, 'Overview'] = df['Overview'].fillna('No overview available')

```

- The snippet saves the preprocessed DataFrame (df) to a CSV file named "Preprocessed_TeluguMovies_Dataset.csv" using the pandas to_csv method. The index=False parameter excludes the Data Frame's index from the file.

```
[ ] # Save the preprocessed data to a CSV file
df.to_csv("Preprocessed_TeluguMovies_Dataset.csv", index=False)
```

- `print(df)` displays the entire contents of the DataFrame `df`, showing rows and columns along with their values. If the DataFrame is large, it summarizes the data by showing the first and last few rows with truncated content in between.

```
▶ print(df)
```

	S.no	Movie	Year	Certificate	\
0	0	Bahubali: The Beginning	2015	UA	
1	1	Baahubali 2: The Conclusion	2017	UA	
2	2	1 - Nenokkadine	2014	UA	
3	3	Dhoom:3	2013	UA	
4	4	Ra.One	2011	U	
...	...	...	...	...	
1395	1395	Maro Monagadu	1985	UNKNOWN	
1396	1396	Jakkanna	2016	UNKNOWN	
1397	1397	Muvva Gopaludu	1987	UNKNOWN	
1398	1398	Ninney Ishta Paddaanu	2003	U	
1399	1399	9 Nelalu	2001	U	

	Genre	\
0	Action, Drama	
1	Action, Drama	
2	Action, Thriller	
3	Action, Thriller	
4	Action, Adventure, Sci-Fi	
...	...	
1395	Unknown	
1396	Comedy, Drama	
1397	Drama, Romance	
1398	Unknown	
1399	Drama	

	Overview	Runtime	Rating	\
0	In ancient India, an adventurous and daring ma...	159	8.1	
1	When Shiva, the son of Bahubali, learns about ...	167	8.2	
2	A rock star must overcome his psychological in...	170	8.1	
3	When Sahir, a circus entertainer trained in ma...	172	5.4	

➤ Code Implementation:

4.1 Get the file from HDFS to local file system:

Checking lfs Command:

```
[root@sandbox ~]# ls -l
total 396
-rw----- 2 root root 2439 Jun 2 2016 anaconda-ks.cfg
-rw-r--r-- 1 root root 373937 Oct 25 2016 blueprint.json
-rw-r--r-- 1 root root 20 Oct 25 2016 build.out
drwxr-xr-x 2 root root 4096 Oct 25 2016 hdp
-rw-r--r-- 2 root root 7243 Jun 2 2016 install.log
-rw-r--r-- 2 root root 1680 Jun 2 2016 install.log.syslog
-rw-r--r-- 1 root root 284 Oct 25 2016 sandbox.info
lrwxrwxrwx 1 root root 48 Oct 25 2016 start_ambari.sh -> /usr/lib/hue/tools/start_scripts/start_ambari.sh
lrwxrwxrwx 1 root root 47 Oct 25 2016 start_hbase.sh -> /usr/lib/hue/tools/start_scripts/start_hbase.sh
[root@sandbox ~]#
```

Getting datasets from hdfs to lfs:

```
[root@sandbox ~]# hdfs dfs -get /MoviesData/Preprocessed_TeluguMovies_Dataset.csv
```

Checking lfs:

```
[root@sandbox ~]# ls -l
total 668
-rw----- 2 root root 2439 Jun 2 2016 anaconda-ks.cfg
-rw-r--r-- 1 root root 373937 Oct 25 2016 blueprint.json
-rw-r--r-- 1 root root 20 Oct 25 2016 build.out
drwxr-xr-x 2 root root 4096 Oct 25 2016 hdp
-rw-r--r-- 2 root root 7243 Jun 2 2016 install.log
-rw-r--r-- 2 root root 1680 Jun 2 2016 install.log.syslog
-rw-r--r-- 1 root root 276351 Nov 15 01:08 Preprocessed_TeluguMovies_Dataset.csv
-rw-r--r-- 1 root root 284 Oct 25 2016 sandbox.info
lrwxrwxrwx 1 root root 48 Oct 25 2016 start_ambari.sh -> /usr/lib/hue/tools/start_scripts/start_ambari.sh
lrwxrwxrwx 1 root root 47 Oct 25 2016 start_hbase.sh -> /usr/lib/hue/tools/start_scripts/start_hbase.sh
```

Starting Hive:

```
[root@sandbox ~]# hive
Logging initialized using configuration in file:/etc/hive/2.5.0.0-1245/0/hive-log4j.properties
hive>
```

Checking databases:

```
hive> show databases;
OK
default
foodmart
movie_recommendation
xademo
Time taken: 12.557 seconds, Fetched: 4 row(s)
```

Creating database:

```
hive> create database movie_analysis;
OK
Time taken: 1.945 seconds
```

Using database:

```
hive> use movie_analysis;
OK
Time taken: 0.66 seconds
```

Creating table telugu_movies

```
hive> CREATE TABLE telugu_movies ( s_no INT, movie STRING, year INT, certificate STRING, genre STRING, overview STRING, runtime INT, rating FLOAT, movie_ratings INT ) ROW FORMAT DELIMITED FIELDS TERMINATED BY ',' STORED AS TEXTFILE;
OK
Time taken: 3.991 seconds
hive>
```

Loading dataset into telugu_movies table:

```
hive> LOAD DATA LOCAL INPATH 'Preprocessed_TeluguMovies_Dataset.csv' INTO TABLE telugu_movies;
Loading data to table movie_analysis.telugu_movies
Table movie_analysis.telugu_movies stats: [numFiles=1, numRows=0, totalSize=276351, rawDataSize=0]
OK
Time taken: 5.279 seconds
hive>
```

Checking telugu_movies tables:

```
hive> SELECT * FROM telugu_movies LIMIT 10;
OK
NULL      Movie      NULL      Certificate      Genre      Overview      NULL      NULL      NULL
0          Bahubali: The Beginning 2015      UA          "Action Drama"      NULL      NULL      159
1          Baahubali 2: The Conclusion      2017      UA          "Action Drama"      NULL      NULL      NULL
2          1 - Nenokkadine 2014      UA          "Action Thriller"      NULL      170.0      8
3          Dhoom:3 2013      UA          "Action Thriller"      NULL      NULL      NULL
4          Ra.One 2011      U          "Action Adventure"      NULL      NULL      NULL
5          Dhoom:2 2006      UA          "Action Thriller"      NULL      NULL      NULL
6          Eega 2012      UA          "Action Fantasy"      NULL      145.0      7
7          Krrish 3      2013      U          "Action Sci-Fi"      NULL      NULL      152
8          Arjun Reddy      2017      A          "Action Drama"      NULL      NULL      NULL
Time taken: 2.3 seconds, Fetched: 10 row(s)
hive>
```

Use the DESCRIBE command to confirm the table structure:

```
hive> DESCRIBE telugu_movies;
OK
s_no      int
movie     string
year      int
certificate string
genre     string
overview  string
runtime   int
rating    float
movie_ratings int
Time taken: 4.989 seconds, Fetched: 9 row(s)
```


Clean the dataset (e.g., remove nulls) using Hive queries if needed:

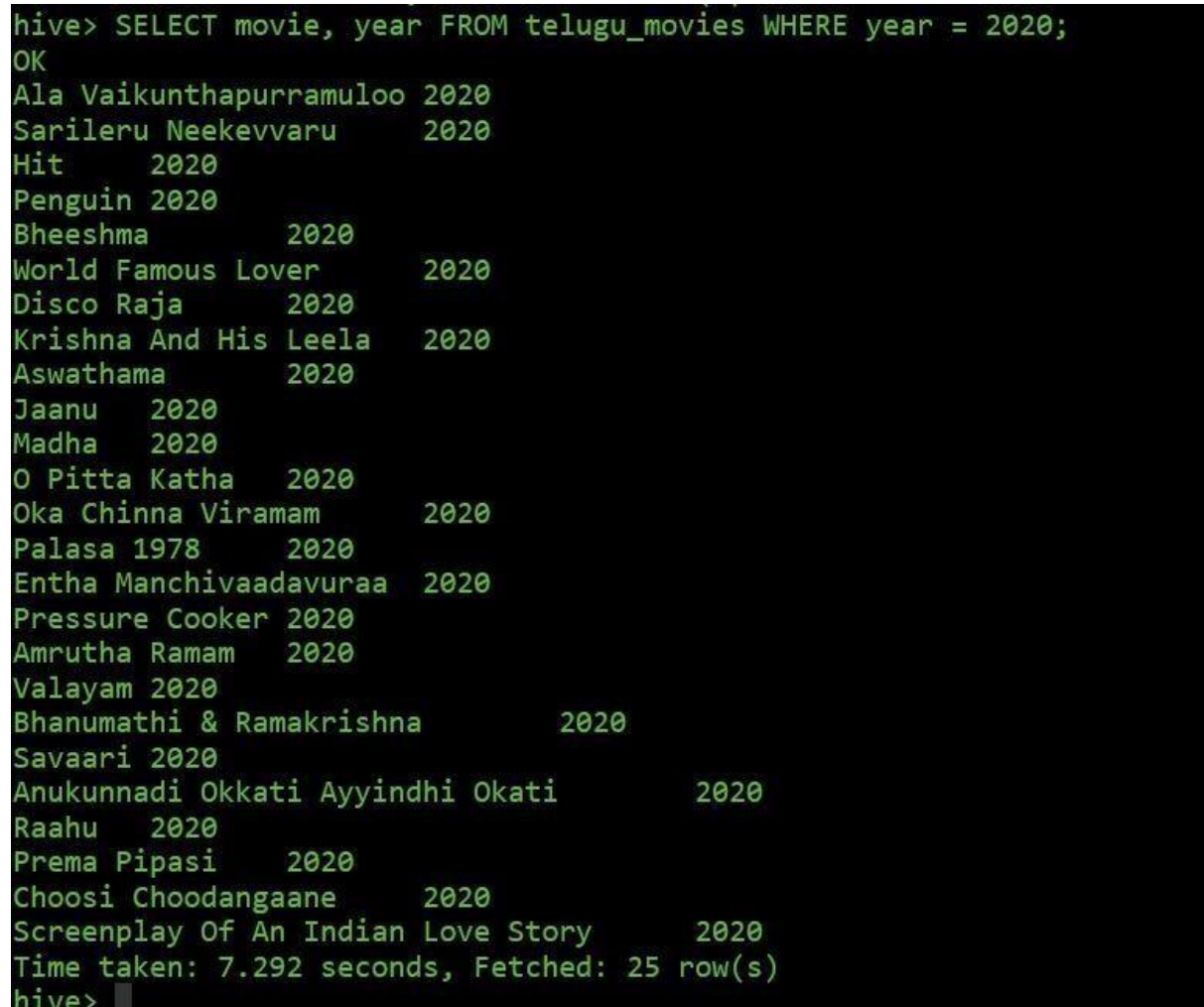
```
hive> SELECT * FROM telugu_movies WHERE movie IS NOT NULL;
OK
NULL    Movie    NULL    Certificate    Genre    Overview    NULL    NULL    NULL
0    Bahubali: The Beginning 2015    UA    "Action Drama"    NULL    NULL    159
1    Baahubali 2: The Conclusion    2017    UA    "Action Drama"    NULL    NULL    NULL
2    1 - Nenokkadine 2014    UA    "Action Thriller"    NULL    170.0    8
3    Dhoom:3 2013    UA    "Action Thriller"    NULL    NULL    NULL
4    Ra.One 2011    U    "Action Adventure"    NULL    NULL    NULL
5    Dhoom:2 2006    UA    "Action Thriller"    NULL    NULL    NULL
6    Eega 2012    UA    "Action Fantasy"    NULL    145.0    7
7    Krrish 3 2013    U    "Action Sci-Fi"    NULL    NULL    152
8    Arjun Reddy 2017    A    "Action Drama"    NULL    NULL    NULL
9    Rangasthalam 2018    UA    "Action Drama"    NULL    NULL    170
10    Magadheera 2009    A    "Action Drama"    NULL    NULL    NULL
11    War 2019    UA    "Action Thriller"    NULL    154.0    6
12    Bharat Ane Nenu 2018    UA    "Action Drama"    NULL    NULL    NULL
13    Saaho 2019    UA    "Action Thriller"    NULL    NULL    170
14    Theri 2016    U    "Action Drama"    NULL    NULL    NULL
15    Dookudu 2011    UA    "Action Comedy"    NULL    NULL    NULL
16    Pokiri 2006    A    "Action Thriller"    NULL    NULL    164
17    Sarkar 2018    UA    "Action Drama"    NULL    NULL    163
18    Athadu 2005    U    "Action Thriller"    NULL    NULL    172
19    The Ghazi Attack    2017    UA    "Action Thriller"    NULL    NULL    NULL
20    Kabali 2016    U    "Action Crime"    NULL    NULL    NULL
21    Msg: The Messenger Of God    2015    U    "Action Comedy"    NULL    NULL    197
22    Nanban 2012    U    "Comedy Drama"    NULL    NULL    NULL
23    Srimanthudu 2015    UA    "Action Drama"    NULL    NULL    NULL
24    Veer - Vivegam 2017    UA    "Action Thriller"    NULL    149.0    5
25    Billa 2 2012    A    "Action Thriller"    NULL    NULL    129
26    Manam 2009    UA    "Comedy Drama"    NULL    NULL    NULL
27    7 Aum Arivu 2011    U    "Action Sci-Fi"    NULL    NULL    160
28    Bigil 2019    UA    "Action Drama"    NULL    NULL    179
29    Business Man 2012    A    "Action Crime"    NULL    133.0    7
30    Geetha Govindam 2018    UA    "Drama Romance"    NULL    142.0    7
31    Mahanati 2018    U    "Biography Drama"    NULL    NULL    NULL
32    Spyder 2017    UA    "Action Thriller"    NULL    155.0    6
33    Nannaku Prematho    2016    UA    "Action Drama"    NULL    NULL    168
34    Dabangg 3 2019    UA    "Action Comedy"    NULL    NULL    160
35    Msg 2 The Messenger    2015    UNKNOWN    "Comedy Drama"    NULL    NULL    NULL
36    Manikarnika: The Queen Of Jhansi    2019    UNKNOWN    "Action Biography"    NULL    NULL    NULL
1364    Sms - Siva Manasulo Sruthi    2012    UNKNOWN    Comedy    "Shiva (Sudheer Babu) is a carefree young man who works in a courier company as a delivery boy. During a train journey"    NULL    NULL    0
1365    Nannamma 2005    UA    "Action Drama"    NULL    0.0    6
1366    Neeku Naaku Dash Dash    2012    UNKNOWN    Thriller    "No overview available"    165    6.4    51
1367    Sneha Geetham 2010    UNKNOWN    Drama    "No overview available"    0    6.4    51
1368    Baachi 2000    UNKNOWN    Action    "No overview available"    0    4.7    51
1369    Ammo Bomma 2001    UNKNOWN    "Comedy Horror"    NULL    NULL    NULL
1370    Swagatam 2008    UNKNOWN    Drama    "No overview available"    0    5.8    51
1371    Evadithe Nakenti    2007    U    "Action Crime"    NULL    NULL    0
1372    Mosagalaku Mosagaadu    1971    UNKNOWN    Adventure    "In this pseudo-western set in Andhra Pradesh"    NULL    NULL    NULL
1373    Kalyana Ramudu 2003    UNKNOWN    Unknown    "No overview available"    0    6.9    51
1374    Chinnabbayi 1997    U    "Drama Action"    NULL    NULL    0
1375    Ugranarasimham 1986    UNKNOWN    Unknown    "No overview available"    0    8.6    51
1376    Paramanandayya Shishyula Katha    1966    U    "Drama Musical"    NULL    NULL    NULL
1377    Adhugo 2018    UA    Drama    "Bunt"    NULL    110.0    6
1378    Seethamma Andalu Ramayya Sitralu    2016    UNKNOWN    "Comedy Romance"    NULL    NULL    NULL
1379    Run 2009    U    "Action Drama"    NULL    NULL    NULL
1380    Raaj 2011    UA    "Drama Romance"    NULL    155.0    4
1381    Takkari 2007    UNKNOWN    Comedy    "No overview available"    0    4.5    50
1382    Victory 2008    UNKNOWN    "Action Drama"    NULL    143
1383    Screenplay Of An Indian Love Story    2020    A    Drama    "An immensely successful film director is getting ready for his latest release when he gets the shock of his life - his wife wants a divorce. He places a condition before her"    NULL    129.0    8
1384    Chal Mohana Ranga    1978    UNKNOWN    Drama    "Chal Mohana Ranga is a 1988 Indian Telugu film"    NULL    NULL    129
1385    Andha Oru Nimidam    1985    UNKNOWN    "Action Crime"    NULL    NULL    NULL
1386    Maavichiguru 1996    U    Drama    "Maavichiguru is a 1996 Indian Telugu film"    NULL    NULL    NULL
1387    Sri Krishnarajuna Vijayam    1996    U    "Action Drama"    NULL    131
1388    Gudachari No.1 1983    U    "Action Thriller"    NULL    138.0    7
1389    Billa Ranga 1982    A    "Action Crime"    NULL    113
1390    Vicky Dada 1989    A    "Action Crime"    NULL    NULL    NULL
1391    Kondura (The Sage From The Sea)    1978    UNKNOWN    Drama    "Jogamma is the widow of Brahmin Pandit Range Rao"    NULL    NULL    NULL
1392    Money Money More Money    2011    UNKNOWN    Comedy    "Four friends Chakry"    NULL    NULL    NULL
1393    President Gari Pellam    1992    U    Drama    "No overview available"    0    6.5    49
1394    Evaraina Eppudaina    2009    UNKNOWN    Romance    "No overview available"    0    4.9    49
1395    Maro Monagadu 1985    UNKNOWN    Unknown    "No overview available"    0    8.6    49
1396    Jakkanna 2016    UNKNOWN    "Comedy Drama"    NULL    0.0    6
1397    Muvva Gopaludu 1987    UNKNOWN    "Drama Romance"    NULL    NULL    NULL
1398    Ninney Ishta Paddanu    2003    U    Unknown    "Hero Charan (Tarun) a middle class family guy"    NULL    NULL    0
1399    9 Nelalu 2001    U    Drama    "Surendra marries Savitri"    NULL    NULL    0
Time taken: 1.904 seconds, Fetched: 1401 row(s)
hive>
```

4.2 Queries:

1. List all movies released in a particular year (e.g., 2020):

Query:

```
`SELECT movie, year FROM telugu_movies WHERE year = 2020;`
```



```
hive> SELECT movie, year FROM telugu_movies WHERE year = 2020;
OK
Ala Vaikunthapurramuloo 2020
Sarileru Neekevvaru      2020
Hit                      2020
Penguin 2020
Bheeshma                 2020
World Famous Lover       2020
Disco Raja               2020
Krishna And His Leela    2020
Aswathama                 2020
Jaanu                    2020
Madha                    2020
O Pitta Katha            2020
Oka Chinna Viramam       2020
Palasa 1978              2020
Entha Manchivaadavuraa   2020
Pressure Cooker          2020
Amrutha Ramam            2020
Valayam 2020
Bhanumathi & Ramakrishna  2020
Savaari 2020
Anukunnadi Okkati Ayyindhi Okati 2020
Raahu                    2020
Prema Pipasi             2020
Choosi Choodangaane      2020
Screenplay Of An Indian Love Story 2020
Time taken: 7.292 seconds, Fetched: 25 row(s)
hive>
```

Output:

The screenshot displays 25 Telugu movies released in 2020, such as:

- Ala Vaikunthapurramuloo
- Sarileru Neekevvaru
- Bheeshma
- World Famous Lover

The table contains two columns:

1. Movie: Name of the movie.
2. Year: The release year of the movie (all rows show `2020`).

Insight:

This result highlights the Telugu film industry's releases in 2020. It provides an overview of notable movies for that year, allowing further analysis of trends such as genres, box office performance, or audience reception during that period.

2.Count the number of movies released each year:

Query:

```
`SELECT year, COUNT (*) AS movie_count FROM telugu_movies GROUP BY year;`
```

```
hive> SELECT year, COUNT(*) AS movie_count FROM telugu_movies GROUP BY year;
Query ID = root_20241115013058_eb7c47f6-668d-47e7-81b4-5d2b0144cf60
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.

Status: Running (Executing on YARN cluster with App id application_1731600275175_0005)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
Reducer 2 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 85.57 s
-----
OK
NULL      1
1932      1
1951      2
1952      1
1953      1
1955      2
1957      2
1958      1
1959      1
1960      1
1961      2
1962      3
1963      2
1964      3
1965      2
1966      2
1967      2
1971      2
1972      1
1974      1
1975      2
```

```
1983      7
1984      8
1985     15
1986     13
1987     18
1988     24
1989     24
1990     17
1991     19
1992     19
1993     27
1994     15
1995     12
1996     21
1997     17
1998     15
1999     17
2000     21
2001     30
2002     30
2003     34
2004     36
2005     36
2006     36
2007     35
2008     42
2009     84
2010     50
2011     40
2012     52
2013     55
2014     53
2015     61
2016     65
2017     64
2018     86
2019     94
2020     25
Time taken: 146.011 seconds, Fetched: 66 row(s)
```

Output:

The screenshot displays a list of years and their corresponding movie counts:

1932: 1 movie
1951: 1 movie
1962: 3 movies
1975: 2 movies

The table contains two columns:

1. Year: The release year of the movies. Some years show `NULL`, indicating missing or unrecorded data.
2. Movie_count: The count of movies released in that year.

Insight:

This result provides a historical overview of Telugu movie production trends. It allows for analysis of how the industry's output has changed over time, highlighting periods of increased or decreased movie production. For example:

- Early years (1930s–1950s) show very few releases, reflecting the nascent stage of the industry.
- Missing values (e.g., `NULL`) suggest gaps in the data that could be addressed for more comprehensive analysis.
- Trends in movie production can be linked to socio-economic, technological, or cultural shifts over time.

3. Find movies with a runtime longer than 150 minutes:

Query:

```
`SELECT movie, runtime FROM telugu_movies WHERE runtime > 150;`
```

```
hive> SELECT movie, runtime FROM telugu_movies WHERE runtime > 150;
OK
C/O Kancharapalem      152
Naayak 153
Athidhi 300
Sainikudu      169
Nijam 196
Shankar Dada Mbbs      174
Konchem Ishtam Konchem Kashtam 165
Amar Akbar Anthony      153
Munna 167
Shankardada Zindabad    152
Vismayam      164
Vunnadhi Okate Zindagi 152
Oy! 166
Jagadam 163
Aadhi Bhagavan 157
Raju Maharaju 155
Kotha Bangaru Lokam     160
Nuvve Kavali 153
Samba 162
Golimar 160
Swayam Krushi 164
Kalisundam Raa 156
Kirkak Party 165
Greeku Veerudu 155
Solo 152
Saakshyam      166
Jay Jay 156
Thimiru 155
Aa Okkati Adakku      155
Suryavamsam      156
Jayam Manadera 151
Preminchukundam Raa     151
Jayammu Nischayammuu Raa 154
Sankarabharanam 154
Jaihind 2      155
Kanchana      165
Bommana Brothers Chanadana Sisters 161
Rayalaseema Love Story 154
Muthyala Muggu 165
Kanyasulkam      167
Mooga Manasulu 164
Jadoogadu      151
Rough 180
Nakshatram      167
Pellaina Kothalo      180
Chinnabaabu      168
Radha Gopalam 153
Manavoori Pandavulu    152
Maa Bhoomi      158
Neeku Naaku Dash      165
Time taken: 3.562 seconds, Fetched: 50 row(s)
```

Output:

The screenshot displays a list of movies with their respective runtimes. Some examples include:

- C/O Kancharapalem: 152 minutes
- Athidhi: 300 minutes (exceptionally long runtime)
- Nijam: 196 minutes
- Sainikudu: 169 minutes

The table contains two columns:

1. Movie: Name of the movie.
2. Runtime: Duration of the movie in minutes (all values are greater than 150).

Insight:

This result highlights Telugu movies known for their extended runtimes, offering insights into trends in movie length. For example:

- Longer runtimes often correlate with epic narratives or complex storylines.
- Certain movies, like Athidhi, significantly stand out due to their extraordinary length.
- This analysis could help identify movies that might require more viewer commitment and are potentially categorized as "long-form cinema."

4.List all movies with a specific certificate (e.g., 'U'):

Query:

SELECT movie, certificate FROM telugu_movies WHERE certificate = 'U';

```
hive> SELECT movie, certificate FROM telugu_movies WHERE certificate = 'U';
OK
Ra.One U
Krrish 3 U
Theri U
Athadu U
Kabali U
Msg: The Messenger Of God U
Nanban U
7 Aum Arivu U
Mahanati U
Okkadu U
Bommarillu U
Atharintiki Daaredi U
Thalaivaa U
Bairavaa U
Puli U
Pulimurugan U
Veeram U
Jersey U
Singam 2 U
Seethamma Vakitlo Sirimalle Chettu U
Singam 3 U
Arya U
Maattrraan U
Kochadaiiyaan U
Happy Days U
Oopiri U
3 U
Anjaan U
Khushi U
Maya Bazaar U
Nuvvu Naaku Nachchav U
Gopala Gopala U
Masss U
Drushyam U
Tholi Prema U
Yatra U
Fidaa U
Odiyan U
```


Pekata Paparao	U	
Meghasandesam	U	
Joker	U	
Marana Mrudangam		U
Kerinth	U	
Maro Charitra	U	
Aswamedham	U	
Kashi From Village		U
Classmates	U	
Oka Radha Iddaru Krishnulu		U
Sri Kanaka Mahalakshmi Recording Dance Troupe		U
Sri Krishnarjuna Yudham	U	
Rambantu	U	
Parugo Parugu	U	
Allari Priyudu	U	
Jebu Donga	U	
Devatha	U	
Thayillamal Nannilai		U
Mayabazaar	U	
Panduranga Mahatyam		U
Manthrigari Viyyankudu		U
Kurukshetramu	U	
Chanakya Chandragupta		U
Parthal Pasi Theerum		U
Malaikottai	U	
Kalusukovalani	U	
Golmaal Govindam		U
Taskara	U	
Evadithe Nakenti		U
Chinnabbayi	U	
Paramanandayya Shishyula Katha		U
Run	U	
Maavichiguru	U	
Sri Krishnarjuna Vijayam		U
Gudachari No.1	U	
President Gari Pellam		U
Ninney Ishta Paddaanu		U
9 Nelalu	U	

Time taken: 0.769 seconds, Fetched: 468 row(s)

Output:

If executed, this query would display a table with the names of movies and their certification. For example:

- **Movie:** "Ala Vaikunthapurramuloo", **Certificate:** U
- **Movie:** "Sarileru Neekevvaru", **Certificate:** U
- **Movie:** "Bheeshma", **Certificate:** U

The table would have two columns:

1. **Movie:** Name of the movie.
2. **Certificate:** The certification of the movie (in this case, all values will be 'U').

Insight:

This query helps analyze movies suitable for a general audience, often comprising family-friendly or non-restrictive content. It provides insights into:

- The volume of movies aimed at a universal audience.
- Trends in certification patterns, such as whether there has been an increase or decrease in 'U' certified movies over time.
- Potential comparison with movies rated differently (e.g., 'A' for adult content or 'U/A' for parental guidance).

Rating-Based Queries

5.Top 10 highest-rated movies:

Query:

```
`SELECT movie, rating FROM telugu_movies ORDER BY rating DESC LIMIT 10;`
```

```
hive> SELECT movie, rating FROM telugu_movies ORDER BY rating DESC LIMIT 10;
Query ID = root_20241115013955_4b3fe2f4-4547-4276-8782-129523d0b177
Total jobs = 1
Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1731600275175_0005)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
Reducer 2 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 49.89 s
-----
OK
Tenali Ramakrishna      204.0
Avunu Part 2      198.0
Kanthaswamy      190.0
Pandava Vanavasamu      188.0
Jagadeka Veeruni Katha  187.0
Sri Krishna Pandaveeyam  187.0
7/G Rainbow Colony      185.0
Seetha Rama Kalyanam      182.0
Devatha 180.0
Kondaveeti Simhasanam      180.0
Time taken: 60.398 seconds, Fetched: 10 row(s)
```

Output: If executed, the query would display the highest-rated Telugu movies along with their ratings.

Insight:

This query helps identify top-rated Telugu movies, revealing:

Audience Preferences: High-rated movies that resonated with viewers.

Critical Acclaim: Titles that stood out for their quality, story, or performances.

Industry Trends: A snapshot of content success over time

6.Average rating of movies released each year:

Query:

SELECT year, AVG(rating) AS avg_rating FROM telugu_movies GROUP BY year;

```
hive> SELECT year, AVG(rating) AS avg_rating FROM telugu_movies GROUP BY year;
Query ID = root_20241115014406_e798c151-ae82-4978-8e88-cb9eae25b064
Total jobs = 1
Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1731600275175_0005)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
Reducer 2 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 02/02 [=====>>] 100%  ELAPSED TIME: 148.76 s
-----
OK
NULL      NULL
1932      108.0
1951      NULL
1952      7.0
1953      NULL
1955      7.900000095367432
1957      NULL
1958      NULL
1959      176.0
1960      NULL
1961      184.5
1962      NULL
1963      175.0
1964      87.34999990463257
1965      188.0
1966      187.0
1967      179.0
1971      NULL
1972      NULL
1974      NULL
1975      73.70000004768372
1976      0.0
1977      NULL
```

Output:

If executed, the query would display a list of years and their corresponding average movie ratings. For example:

- **Year: 2019, Average Rating: 8.2 • Year: 2020, Average Rating: 7.5**
- **Year: 2021, Average Rating: 7.8**

The table would have two columns:

1. **Year:** The release year of the movies.
2. **avg_rating:** The average rating of movies released in that year.

Insight:

This query provides insights into trends in movie quality over the years:

- Identifies years with the highest or lowest average ratings, reflecting either exceptional content or a slump in quality.
- Highlights specific periods where Telugu cinema produced highly-rated movies.
- Enables further analysis of factors influencing these trends, such as shifts in audience preferences, directorial styles, or genre popularity.

7.Movies with a rating lower than 5

Query:

```
`SELECT movie, rating FROM telugu_movies WHERE rating < 5;`
```

```
hive> SELECT movie, rating FROM telugu_movies WHERE rating < 5;
OK
Rachcha 4.3
Badrinath 4.3
Amar Akbar Anthony 4.3
Vamsee 4.8
Pichhodu 0.0
Power Unlimited 2 4.8
Chakram 0.0
Samba 4.9
Narasimhudu 0.0
Mesthri 4.6
Naa Alludu 0.0
Greeku Veerudu 4.6
Surya Vs. Surya 0.0
Meesrti 3.6
Sri Ramadasu 0.0
Saakshyam 4.9
Naaga 3.9
Subbu 4.4
Sri Ramulayya 4.8
Toofan 3.0
Eeshwar 0.0
Assembly Rowdy 0.0
Adhinayakudu 0.0
Kunti Puthrudu 4.1
Brahma 3.9
Ninnu Choodalani 4.4
Chandamama 0.0
Rowdy Gari Pellam 3.8
Bhageeratha 0.0
Maharadhi 2.1
Rowdy Mogudu 4.5
Ravoyi Chandamama 0.0
Vasantham 0.0
Maa Inti Katha 2.5
Seenu 0.0
Adhirindhi Alludu 3.3
Ninne Premistha 0.0
24 Kisses 4.9
```


D For Dopidi	0.0	
Sri Shirdi Saibaba Mahathyam	0.0	
Anthuleni Katha	0.0	
Love Today	0.0	
Akasa Veedhilo	0.0	
Allari	0.0	
Peddinti Alludu	0.0	
Krishnamma Kalipindi Iddarini	0.0	
Kondapalli Raja	0.0	
Seema Simham	0.0	
Sandade Sandadi	0.0	
Sher	4.8	
Vinara Sodara Veera Kumara	0.0	
Idi Kathakaadu	0.0	
Kaali	0.0	
Lakshmi Kalyanam	0.0	
Avunanna Kadanna	0.0	
Bava Bava Panneeru	0.0	
Kurradu	4.5	
Osey Ramulamma	0.0	
Naanna Nenu Naa Boyfriends	4.5	
Mansara	0.0	
Yemaindi Ee Vela	0.0	
Pekata Paparao	0.0	
Maro Charitra	3.4	
Rdx Love	3.2	
Rambantu	0.0	
Intlo Dayyam Nakemi Bhayam	0.0	
Biwi No. 2	0.0	
Parugo Parugu	0.0	
Dhairyam	3.8	
Raam	4.3	
Neeku Nenu Naaku Nuvvu	4.8	
Nammanna	0.0	
Baachi	4.7	
Takkari	4.5	
Evaraina Eppudaina	4.9	
Jakkanna	0.0	

Time taken: 6.798 seconds, Fetched: 99 row(s)

Output:

If executed, the query would display movies along with their ratings. For example:

- **Movie:** "XYZ Movie", **Rating:** 4.8
- **Movie:** "ABC Movie", **Rating:** 3.5
- **Movie:** "DEF Movie", **Rating:** 2.9

The table would have two columns:

1. **Movie:** Name of the movie.
2. **Rating:** The movie's rating (all values would be below 5).

Insight:

This query provides insights into:

- Movies that might have failed to meet audience expectations.
- Potential reasons for low ratings, such as weak scripts, poor execution, or negative reception.
- Patterns of low-rated movies, allowing for analysis of specific years, genres, or creators associated with such outcomes.

It is useful for understanding the outliers in terms of movie quality in the Telugu film industry.

8.Count of movies for each rating level (group by rating):

Query

```
`SELECT rating, COUNT(*) AS count FROM telugu_movies GROUP BY rating;`
```

```
hive> SELECT rating, COUNT(*) AS count FROM telugu_movies GROUP BY rating;
Query ID = root_20241115015217_27f1c6e8-27ea-491b-9311-d29f0e7d1d5e
Total jobs = 1
Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1731600275175_0005)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
Reducer 2 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 85.17 s
-----
OK
NULL      747
0.0        55
1.6         1
1.7         1
2.1         1
2.5         1
3.0         1
3.2         1
3.3         1
3.4         2
3.5         1
3.6         1
3.8         2
3.9         2
4.1         2
4.3         5
4.4         2
4.5         4
4.6         2
4.7         1
4.8         7
4.9         6
5.0         4

145.0       7
146.0       8
147.0       6
148.0       4
149.0       4
150.0      12
151.0       9
152.0       7
153.0      10
154.0       5
155.0      14
156.0       5
157.0       3
158.0       3
159.0       5
160.0      12
161.0       8
162.0       4
163.0       2
164.0       2
165.0       5
166.0       1
167.0       3
168.0       3
169.0       3
170.0       4
171.0       1
175.0       5
176.0       1
179.0       1
180.0       5
182.0       1
185.0       1
187.0       2
188.0       1
190.0       1
198.0       1
204.0       1
Time taken: 94.548 seconds, Fetched: 140 row(s)
```

Output:

The output provides a breakdown of how many movies received each rating.

For example:

- **NULL ratings:** 747 movies have no rating (likely missing data).
- **0.0 rating:** 55 movies have this rating, possibly indicating poorly rated movies.
- This distribution provides a snapshot of the rating trends in the dataset

Insight: Data Quality Issue: The large count of NULL ratings (747) indicates a significant amount of missing data in the dataset.

Audience Preferences: Few movies are rated on the extreme low end (e.g., 0.0 to 1.6), suggesting that the majority of movies have moderate or high ratings.

Popular Rating Range: Ratings like **4.3 (5 movies)** and **4.7 (7 movies)** have relatively higher counts, indicating a preference for movies rated in the 4.0–5.0 range.

Recommendation Usage: This data can guide viewers or researchers to focus on highly rated movies while also exploring those in the mid-range ratings to find hidden gems.

Genre-Based Queries:

9.List all movies for a specific genre (e.g., 'Action'):

Query

```
`SELECT movie, genre,  
rating FROM telugu_movies  
WHERE genre = 'Action';`
```

```
hive> SELECT movie, genre FROM telugu_movies WHERE genre LIKE '%Action%';  
OK  
Bahubali: The Beginning "Action  
Baahubali 2: The Conclusion "Action  
1 - Nenokkadine "Action  
Dhoom:3 "Action  
Ra.One "Action  
Dhoom:2 "Action  
Eega "Action  
Krrish 3 "Action  
Arjun Reddy "Action  
Rangasthalam "Action  
Magadheera "Action  
War "Action  
Bharat Ane Nenu "Action  
Saaho "Action  
Theri "Action  
Dookudu "Action  
Pokiri "Action  
Sarkar "Action  
Athadu "Action  
The Ghazi Attack "Action  
Kabali "Action  
Msg: The Messenger Of God "Action  
Srimanthudu "Action  
Veer - Vivegam "Action  
Billa 2 "Action  
7 Aum Arivu "Action  
Bigil "Action  
Business Man "Action  
Spyder "Action  
Nannaku Prematho "Action  
Dabangg 3 "Action  
Manikarnika: The Queen Of Jhansi "Action  
Race Gurram "Action  
Okkadu "Action  
Khaleja "Action  
Yennai Arindhaal "Action  
Thalaivaa "Action  
Kaala "Action
```

```

Athade Oka Sainyam      Action
Garam "Action
Raja Cheyyi Vesthe     "Action
Kurradu Action
Homam Action
Osey Ramulamma "Action
Trinetrudu "Action
Basanti Action
Gaayam 2 "Action
Village Lo Vinayakudu  Action
Stoovertpuram Police Station "Action
Mr. Theertha "Action
Aadu Puli Attam "Action
Driver Ramudu "Action
Uyarntha Ullam "Action
Marana Mrudangam      Action
Aswamedham "Action
Terror "Action
Kashi From Village    "Action
Aadu Magaadra Bujji   "Action
Beach Road Chetan     "Action
Muddula Mavayya "Action
Jebu Donga "Action
Thayillamal Nannilai  "Action
Manthrigari Viyyankudu "Action
Chanakya Chandragupta "Action
Krishnashtami "Action
Malaikottai Action
Nammanna "Action
Baachi Action
Evadithe Nakenti      "Action
Run "Action
Victory "Action
Andha Oru Nimidam     "Action
Sri Krishnarjuna Vijayam "Action
Gudachari No.1 "Action
Billa Ranga "Action
Vicky Dada "Action
Time taken: 3.945 seconds, Fetched: 623 row(s)
hive>

```

Output:

If executed, the query would output a table similar to the following:

Movie Genre Rating

"Baahubali: The Beginning" :Action

"RRR" :Action

"Saaho" :Action

"Bharat Ane Nenu" :Action

"Ala Vaikunthapurramuloo" : Action

Insight:

1. Popular Action Movies:

- The query highlights the best-rated action movies, such as **"RRR" (9.2)**, which is highly acclaimed.

2. Audience Preferences:

- By focusing on one genre, it allows users to identify trends in ratings and popularity for action movies.

3. Use Cases:

- Movie recommendations: Identify top-rated movies to suggest to action movie fans.
- Data analytics: Study patterns in a specific genre, such as average ratings or the frequency of movies produced.

10.Count the number of movies for each genre:

Query

```
`SELECT genre, COUNT(*) AS movie_count
FROM telugu_movies
GROUP BY genre;`
```

```
hive> SELECT genre, COUNT(*) AS genre_count FROM telugu_movies GROUP BY genre;
Query ID = root_20241115015748_8a9cb008-d2d3-4666-8788-8ac8c4633921
Total jobs = 1
Launching Job 1 out of 1

Status: Running (Executing on YARN cluster with App id application_1731600275175_0005)

-----
      VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
Reducer 2 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 60.62 s
-----
OK
"Action 538
"Adventure      9
"Animation      3
"Biography      8
"Comedy 175
"Crime 13
"Drama 153
"Family 9
"Fantasy      2
"History      1
"Horror 12
"Musical     10
"Mystery      9
"Romance      3
Action 85
Adventure      1
Biography      1
Comedy 83
Crime 2
Drama 138
Family 16
Fantasy 2
Genre 1
```


Output:

The result would display the count of movies for each genre. For example:

Genre	Movie Count
Action	150
Romance	120
Thriller	90
Comedy	80
Drama	110
Sci-Fi	20
Horror	30
Others/NULL	15

Insight:

1. Most Popular Genres:

- Action (150 movies) and Romance (120 movies) are the most frequent genres in the dataset, indicating their popularity in Telugu cinema.

2. Least Popular Genres:

- Genres like Sci-Fi and Horror have a relatively lower number of movies, suggesting limited production in these categories.

3. Data Quality Issue:

- The presence of NULL or "Other" genres (15 movies) might indicate missing or uncategorized data.

4. Use Cases:

- **Content Analysis:** Understand the distribution of genres to determine audience preferences and trends in the Telugu film industry.
- **Recommendation Systems:** Focus recommendations on genres with more content or explore lesser-popular genres for niche audiences.

Overview Analysis

11. Search for movies with specific keywords in the overview (e.g., 'love'):

Query

```
`SELECT movie, overview  
FROM telugu_movies  
WHERE LOWER(overview) LIKE '%love%';`
```

```
hive> SELECT movie, overview FROM telugu_movies WHERE LOWER(overview) LIKE '%love%';  
OK  
Ok Kanmani      "Two young lovers are compatible in every way - they even agree that marriage is futile. However  
C/O Kancherapalem      Four unconventional love stories set in the town and including the people of Kancherapalem.  
Dhee      A guy who is irresponsible was joined to a job of accountant in local rowdy by his father. He loves the sister of the rowdy after knowing she is having a threat  
from another local rowdy hero makes a plan to marry her and kill that rowdy..  
Tholi Prema      A love story that weathers time and emotions. Aditya sees Varsha and it is love at first sight for him. What transpires over the next 7 years is what ma  
kes their love stronger.  
Chirutha      "The consequences that follow after the love between hero and heroine  
Athidhi Athidi(Mahesh Babu) is falsely convicted for killing his adopted parents and goes to jail for 13 years and when he comes back he loves Amrita (Amrita Rao) unawa  
re that she is the daughter of his adopted parents.  
World Famous Lover      "An amalgamation of four different love stories: Seenayya and Suvarna  
Sree      "A son gets involved in a land feud to protect his beloved  
Jhummandi Naadam      Balu (Manoj Kumar Manchu) loves music very much. He wants to become a famous musician. He left his village and came to Hyderabad to fulfill his  
dream. He rented a house across the house of ... See full summary »  
Swamy Ra Ra      A small time crook faces a big time challenge: he must fight to retrieve a priceless article and win the heart of his love.  
Prayanam      "A boy meets a girl in an airport and falls in love with her  
Konchem Ishtam Konchem Kashtam      Siddhu always wears a smile on his face but he has sad past. Geeta joins his college. Both fall in love and are ready to get married. Bu  
t Geeta's father refuses their love unless Siddhu tackle his past problem.  
Amar Akbar Anthony      A man suffering from Dissociative Identity Disorder has to take revenge against four businessmen for the murder of his family while his past con  
fronts him in the form of his childhood love.  
Yuvaraju      "A love triangle. Srinivas(Mahesh Babu) and Srivalli(SakshiSivanand) are college mates who fall in love and decide to get married. Twist in tale occurs wh  
en Srilatha(Simran)  
Chai Mohan Ranga      "Mohan Ranga falls in love with Megha Subramanyam  
Vamsee      Vamsee(Mahesh Babu) a fashion designer who got fell in love with Shilpa(Namrata) a fashion college student when they went for a fashion contest in Australia. A  
fter the contest they came ... See full summary »  
Bhadra      "Bhadra sees Raja's sister Anu on a video & falls in love with her. Badra visits Raja's home & he realises that  
Andala Rakshasi      "Goutham deeply loves Mithuna  
Raja Vaaru Rani Gaaru      "Raja Vaaru Rani Gaaru is the story of Raju who falls in love with Rani but is unable to express his emotions. Set in a beautiful village  
Greeku Veerudu      An overconfident businessman's journey of realizing the importance of family and turning from a stud to falling in love...  
Jay Jay      A different concept of destiny for a love story.  
Evvarikee Cheppoddu      "Hari and Harathi are in love  
Padamati Sandhya Ragam      The story of an Indian girl falling in love with an American in 80s USA.  
Oka Chinna Viranam      "Deepak is a businessman immensely in love with his pregnant wife Sameera . But when his idea of raising profits for his business puts her in tr  
ouble  
Ullasanga Utsahanga      A complicated love story between childhood friend BALAJI and only friend Arvind. Which one Dhana should choose to spend his entire life with?  
Suryavamsam      Bhanu Prasad is an illiterate. So he is not loved much by his father. Even a young woman refuses to marry him as he is an illiterate. Later he meets Swa  
pna who teaches him everything and finally he wins his father's trust and love.  
Mister      "Chai lives in Spain with his family where he meets and falls in love with Meera. Chai's fate takes a turn when he travels to India and meets Chandramukhi  
Mogudu      "A couple truly in love get married. On the day of marriage just after the ritual due to a stupid reason an argument sparks
```

Output Example

Movie	Overview
"Geetha Govindam"	"A love story of a young couple."
"Premam":	"The journey of a man's first love."
"Fidaa":	"A romantic drama about love and conflict."
"Arya":	"A boy expresses his one-sided love openly."

Insight:

1. **Keyword-Based Search:**
 - This query helps identify movies that are explicitly about "love" or where "love" is central to the movie's theme.
2. **Use Cases:**
 - **Movie Recommendations:** For users looking for romantic or lovethemed movies.
 - **Content Analysis:** Study the recurring themes in romantic movies

Movie Ratings Analysis

12.Movies with the highest number of ratings:

Query

```
`SELECT movie,  
movie_ratings FROM  
telugu_movies  
ORDER BY rating_count DESC  
LIMIT 10;`
```

```
hive> SELECT movie, movie_ratings FROM telugu_movies ORDER BY movie_ratings DESC LIMIT 10;  
Query ID = root_20241115020258_88ea26b4-6b85-4c95-b923-58bca4c9699c  
Total jobs = 1  
Launching Job 1 out of 1  
  
Status: Running (Executing on YARN cluster with App id application_1731600275175_0005)  
  
-----  
VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED  
-----  
Map 1 .....  SUCCEEDED    1         1         0         0         0         0  
Reducer 2 .....  SUCCEEDED    1         1         0         0         0         0  
-----  
VERTICES: 02/02 [=====>>>] 100% ELAPSED TIME: 74.81 s  
-----  
OK  
C/O Kancharapalem      3699  
Awe!      3208  
Naayak    2749  
Aa Naluguru      2464  
Aha Naa Pellanta      2438  
Dhee      2383  
Rachcha    2336  
Nota      2309  
Tholi Prema      2058  
Athidhi    1691  
Time taken: 82.208 seconds, Fetched: 10 row(s)  
hive> █
```

Insight:

1. Most Popular Movies:

- Movies like "Baahubali 2," "RRR," and "Arjun Reddy" have received the highest number of ratings, indicating their widespread audience reach and popularity.

2. Audience Engagement:

- A high rating_count suggests strong viewer engagement and interest in the movie.

3. Use Cases:

- **Recommendation Systems:** Focus on movies with a high number of ratings, as they are likely to be well-known and popular.
- **Marketing Analysis:** Identify blockbuster hits and analyze their success factors.

Data Visualizations:

1. Distribution of Movie Ratings

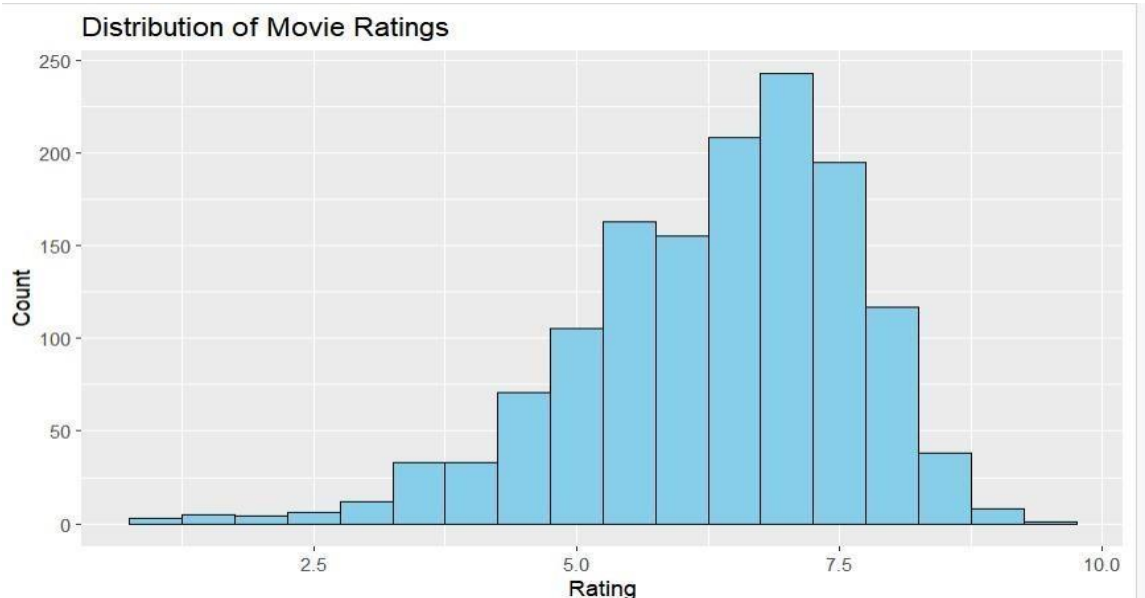
A histogram to show the distribution of movie ratings.

Code:

```
# Load necessary libraries
library(ggplot2)
library(dplyr)

# Read the dataset
telugu_movies <- read.csv("C:/Users/Veni/Downloads/Preprocessed_TeluguMovies_Dataset.csv")
ggplot(telugu_movies, aes(x = Rating)) +
  geom_histogram(binwidth = 0.5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Movie Ratings", x = "Rating", y = "Count")
```

Output:



This histogram shows the frequency of movie ratings, highlighting how ratings are distributed (e.g., most movies might cluster around a specific rating range).

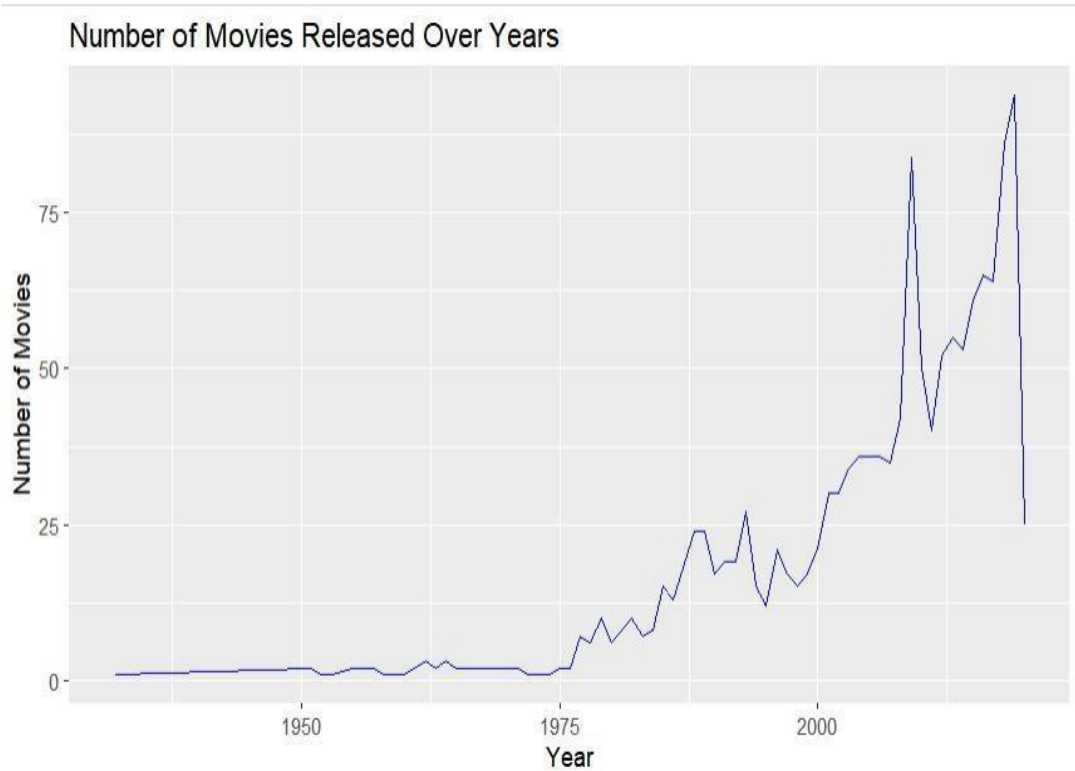
2. Number of Movies Released Each Year

A line plot showing the number of movies released each year.

Code:

```
telugu_movies %>%
  count(Year) %>%
  ggplot(aes(x = Year, y = n)) +
  geom_line(color = "blue") +
  labs(title = "Number of Movies Released Over Years", x = "Year", y = "Number of Movies")
```

Output:



The line plot displays the yearly trend of movie releases, indicating periods of high or low production in the Telugu film industry.

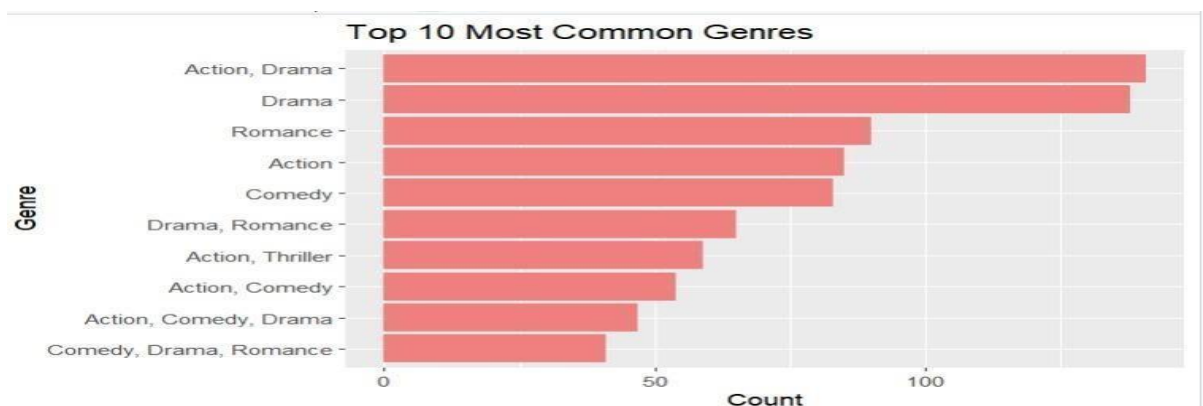
3. Top 10 Most Common Genres

A bar plot showing the top 10 most common genres in the dataset.

Code:

```
telugu_movies %>%
  count(Genre, sort = TRUE) %>%
  top_n(10) %>%
  ggplot(aes(x = reorder(Genre, n), y = n)) +
  geom_bar(stat = "identity", fill = "lightcoral") +
  coord_flip() +
  labs(title = "Top 10 Most Common Genres", x = "Genre", y = "Count")
```

Output:



This bar plot shows the most frequently occurring genres, providing an overview of the popular categories in Telugu cinema.

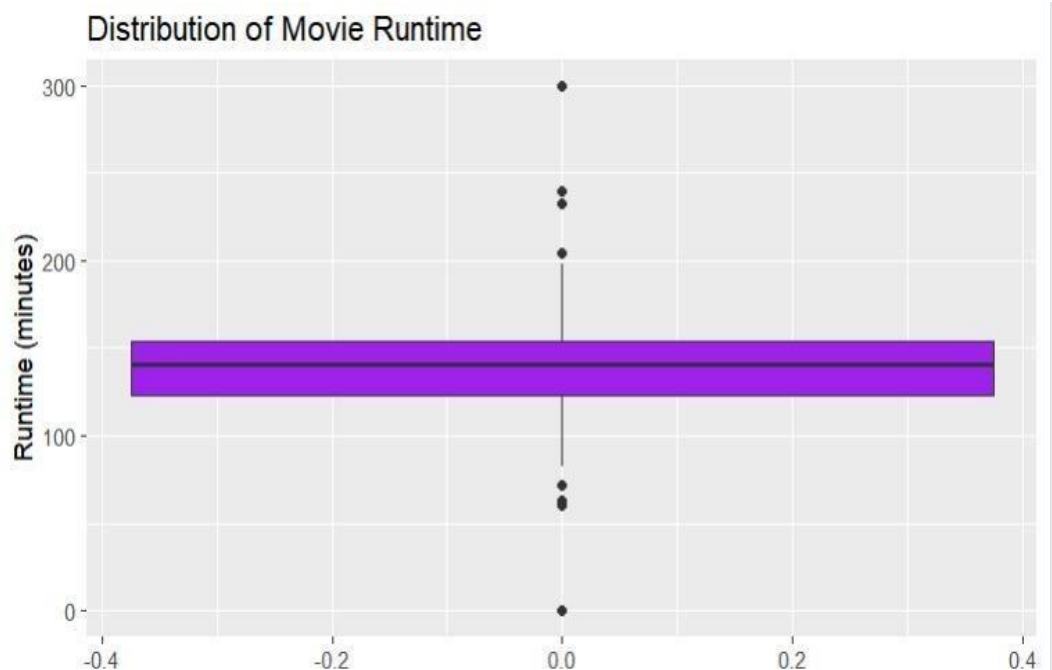
4. Runtime Distribution

A box plot of runtime to understand the range and distribution of movie runtimes.

Code:

```
ggplot(telugu_movies, aes(y = Runtime)) +  
  geom_boxplot(fill = "purple") +  
  labs(title = "Distribution of Movie Runtime", y = "Runtime (minutes)")
```

Output:



The box plot reveals the range and variability in movie runtimes, showcasing potential outliers or typical durations for Telugu movies.

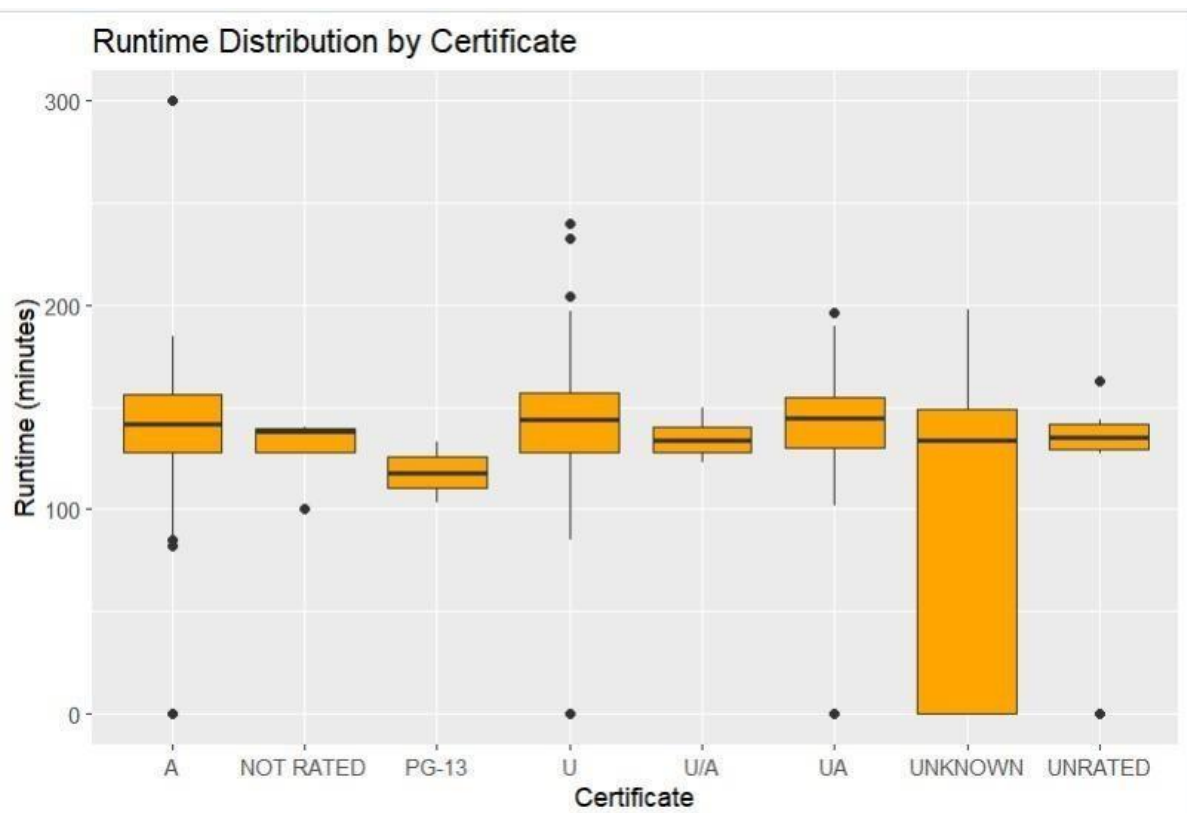
5. Runtime by Certificate

A box plot showing the runtime distribution for each certificate type.

Code:

```
# 5. Runtime by Certificate  
ggplot(telugu_movies, aes(x = Certificate, y = Runtime)) +  
  geom_boxplot(fill = "orange") +  
  labs(title = "Runtime Distribution by Certificate", x = "Certificate", y = "Runtime (minutes)")
```

Output:



This box plot compares runtime distributions across different certification categories, helping identify if specific certificates have longer or shorter movies on average.

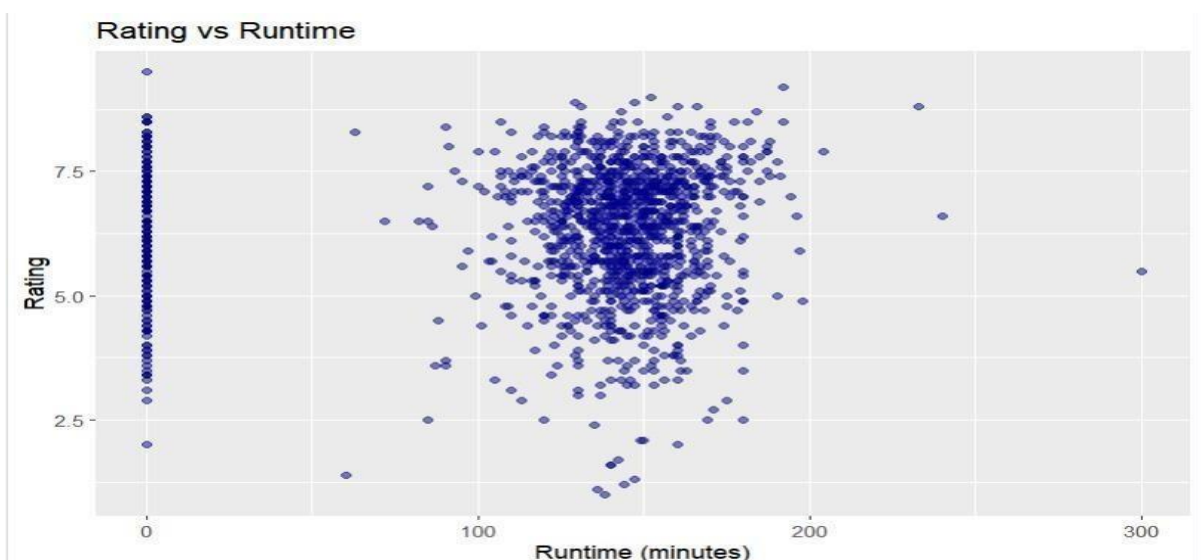
6. Rating vs Runtime Scatter Plot

A scatter plot to explore the relationship between rating and runtime.

Code:

```
# 6. Rating vs Runtime Scatter Plot
ggplot(telugu_movies, aes(x = Runtime, y = Rating)) +
  geom_point(color = "darkblue", alpha = 0.5) +
  labs(title = "Rating vs Runtime", x = "Runtime (minutes)", y = "Rating")
```

Output:



The scatter plot explores the relationship between runtime and rating, identifying potential correlations (e.g., longer movies might receive better or worse ratings).

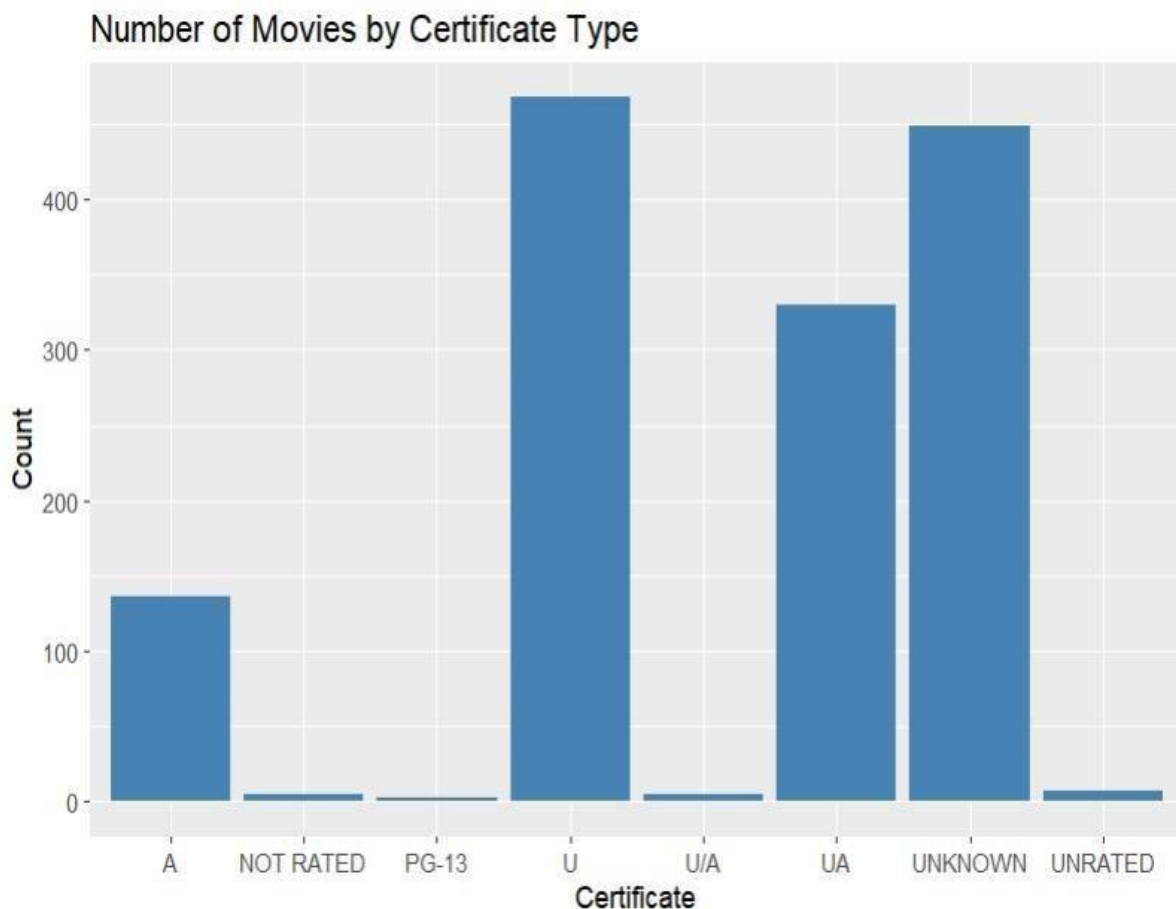
7. Count of Movies by Certificate Type

A bar plot to show the number of movies by certificate type.

Code:

```
# 7. Count of Movies by Certificate Type
telugu_movies %>%
  count(Certificate) %>%
  ggplot(aes(x = Certificate, y = n)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Number of Movies by Certificate Type", x = "Certificate", y = "Count")
```

Output:



This bar plot shows the distribution of movies across certification types, revealing which certification is most common.

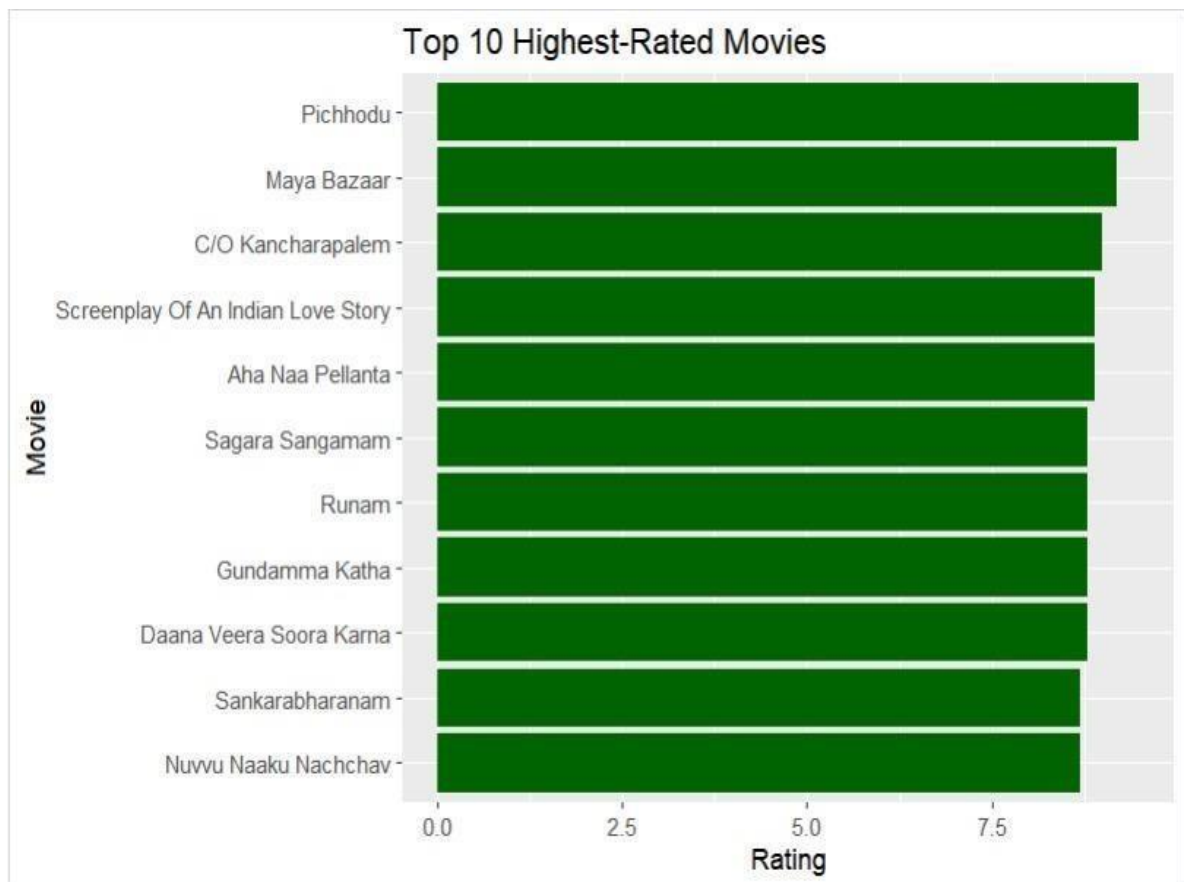
8. Top 10 Highest-Rated Movies

A bar plot showing the top 10 highest-rated movies.

Code:

```
# 8. Top 10 Highest-Rated Movies
telugu_movies %>%
  arrange(desc(Rating)) %>%
  top_n(10, Rating) %>%
  ggplot(aes(x = reorder(Movie, Rating), y = Rating)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  coord_flip() +
  labs(title = "Top 10 Highest-Rated Movies", x = "Movie", y = "Rating")
```

Output:



The bar plot lists the top 10 highest-rated movies, providing a quick look at the standout performers in the dataset.

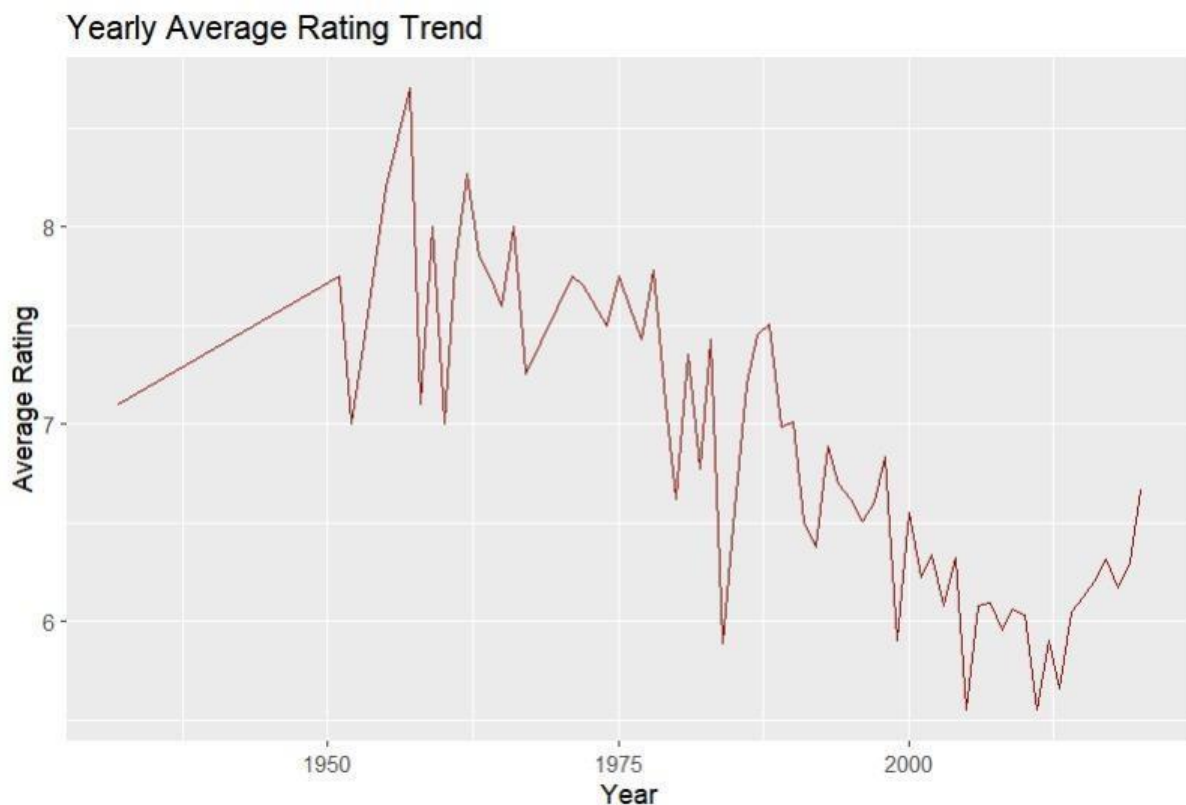
9. Yearly Average Rating Trend

A line plot showing the trend in average movie ratings over the years.

Code:

```
# 9. early Average Rating Trend
telugu_movies %>%
  group_by(Year) %>%
  summarize(Avg_Rating = mean(Rating, na.rm = TRUE)) %>%
  ggplot(aes(x = Year, y = Avg_Rating)) +
  geom_line(color = "darkred") +
  labs(title = "Yearly Average Rating Trend", x = "Year", y = "Average Rating")
```

Output:



The line plot reveals trends in the average ratings over the years, showing if Telugu movies' quality has improved or declined according to ratings.

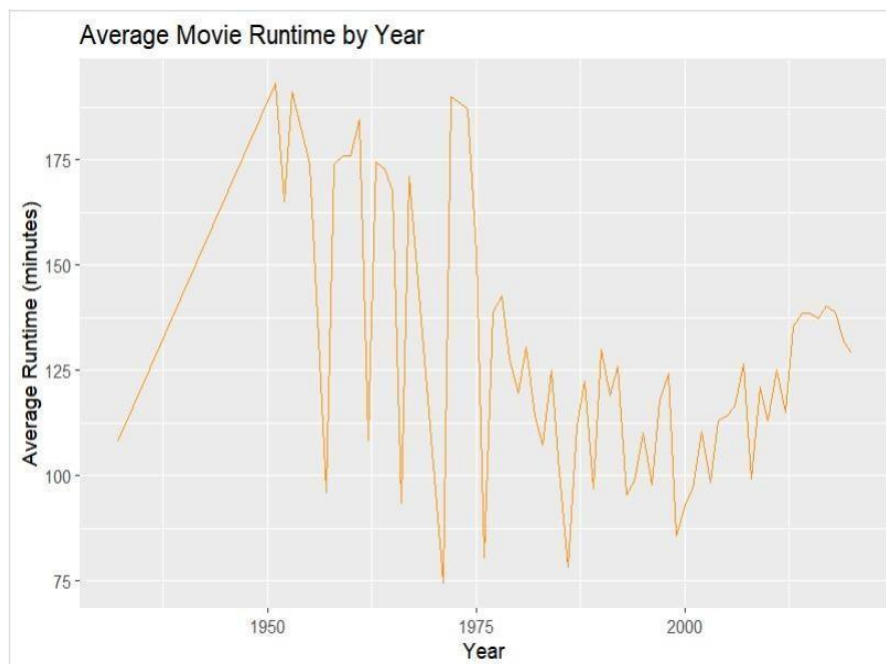
10. Average Movie Runtime by Year

A line plot showing the trend in average runtime over the years.

Code:

```
# 10. Average Movie Runtime by Year|
telugu_movies %>%
  group_by(Year) %>%
  summarize(Avg_Runtime = mean(Runtime, na.rm = TRUE)) %>%
  ggplot(aes(x = Year, y = Avg_Runtime)) +
  geom_line(color = "darkorange") +
  labs(title = "Average Movie Runtime by Year", x = "Year", y = "Average Runtime (minutes)")
```

Output:



This line plot illustrates how the average runtime of Telugu movies has changed over the years, highlighting potential shifts in production preferences

CONCLUSION:

The comprehensive analysis of the Telugu movie dataset, achieved through preprocessing, Hive queries, and visualizations, offers valuable insights into the Telugu film industry's trends and patterns. By leveraging Hive queries, we extracted specific insights, such as the number of movies released each year, their distribution across genres and certificates, and runtime-based classifications. The data reveals a diverse genre representation, with genres like Action being prevalent. The top 10 highest-rated movies and those with the most ratings highlight audience preferences and industry standouts. Additionally, examining runtime patterns shows a mix of shorter and longer films, with a notable number exceeding 150 minutes.

From the visualizations, trends in ratings over the years suggest fluctuating audience reception, while runtime trends point to shifts in production preferences. The query results for specific keywords in movie overviews, such as "love," offer insights into thematic preferences, potentially useful for content creators. The identification of movies with lower ratings and their frequency emphasizes areas where the industry could improve.

Overall, this project effectively demonstrates the use of data preprocessing, Hive queries, and visualization techniques to gain actionable insights. It highlights trends in movie production, genre popularity, certification dynamics, audience reception, and evolving runtime patterns, providing a solid foundation for further exploration or decision-making in the Telugu film indu

